

Project Initialization and Planning Phase

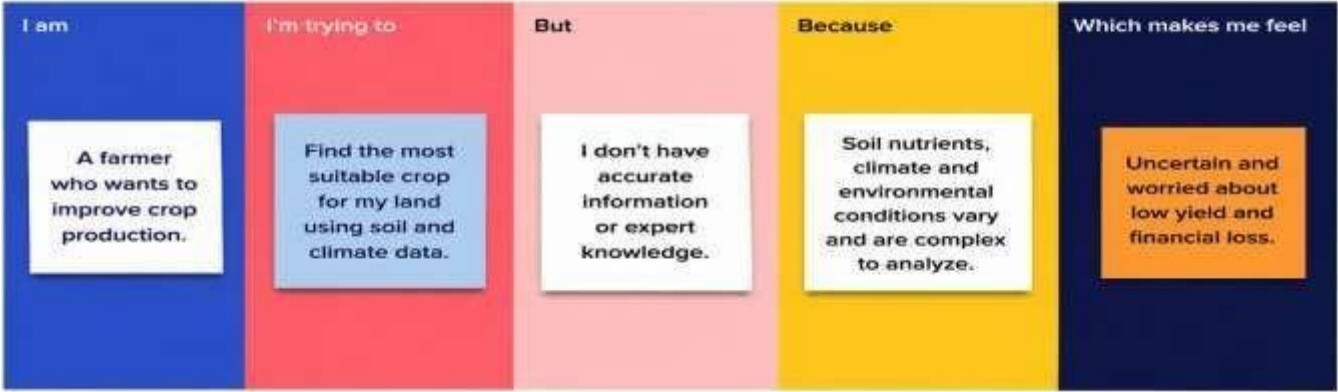
Date	4 July 2026
Team ID	SWUID2026211863
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A farmer	Select the best crop for my land.	I don't know which crop is suitable for my soil and climate conditions.	I have limited knowledge about soil nutrients and weather impact on crops.	Confused and worried about crop yield.

Developed an intelligent crop recommendation system that analyzes soil nutrients (N, P, K), temperature, humidity, pH, and rainfall using machine learning to recommend the most suitable crop, thereby improving agricultural productivity, resource efficiency, and sustainable farming practices.

Example:



PS-2	An agricultural researcher	Analyze crop and environmental data.	It is difficult to identify patterns from large agricultural datasets.	Manual analysis is timeconsuming and complex.	Frustrated and less productive.
PS-3	An agricultural policymaker	Develop strategies to improve agricultural productivity.	I don't have accurate datadriven insights for decisionmaking.	Crop production depends on multiple environmental factors.	Concerned about making effective policies.